

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & 0 \end{bmatrix}$$

$$M_x = M_{px}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = M_{dxz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = -T_{py}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ M_{yyy} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 2M_{fax}^{(1)} + 3M_{px}^{(1)}$$

$$M_{yyx} = -M_{fax}^{(1)} + M_{fbx}^{(1)} + M_{px}^{(1)}$$

$$M_{zzx} = -M_{fax}^{(1)} - M_{fbx}^{(1)} + M_{px}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{yyx} = 2M_{px}^{(2)} + 2T_{dyz}^{(1)}$$

$$M_{zzx} = 2M_{px}^{(2)} - 2T_{dyz}^{(1)}$$

$$\frac{M_{zzx}}{2} = M_{px}^{(2)} - T_{dyz}^{(1)}$$

$$\frac{M_{yyx}}{2} = M_{px}^{(2)} + T_{dyz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zzz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{zzz} = M_{px}^{(3)} + T_{dyz}^{(2)}$$

$$M_{xyy} = -M_{px}^{(3)} + T_{dyz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{gay}^{(1)} - M_{gby}^{(1)}$$

$$M_{yyzx} = M_{dxz}^{(1)} + 2M_{gby}^{(1)}$$

$$M_{zzzx} = M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{gay}^{(1)} - M_{gby}^{(1)}$$

$$M_{yzxy} = M_{dxz}^{(2)} + 2M_{gby}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = M_{dxz}^{(3)} + 2T_{fay}^{(1)} + 2T_{fby}^{(1)} - 2T_{py}^{(1)}$$

$$M_{yyzx} = M_{dxz}^{(3)} - 4T_{fby}^{(1)}$$

$$M_{zzzx} = M_{dxz}^{(3)} - 2T_{fay}^{(1)} + 2T_{fby}^{(1)} + 2T_{py}^{(1)}$$

$$M_{yzxy} = 4T_{fay}^{(1)} + T_{py}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = 2M_{dxz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$M_{yzxy} = T_{py}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & 0 \\ 0 & M_{yyzx} & 0 \\ 0 & M_{zzzx} & 0 \\ 0 & 0 & M_{yzxy} \\ 0 & 0 & 0 \\ M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= 2M_{dxz}^{(5)} - T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(3)} \\
M_{yyzx} &= 2T_{fay}^{(2)} + T_{py}^{(3)} + 2T_{py}^{(4)} \\
M_{zzzx} &= -2M_{dxz}^{(5)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(3)} \\
M_{yzxy} &= M_{dxz}^{(5)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(4)} \\
M_{xyyz} &= -M_{dxz}^{(5)} - T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(4)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ M_{xxxx} & M_{xyyy} & M_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxy} &= -M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(6)} \\
M_{xxxx} &= 2M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{xyyy} &= 2T_{fay}^{(3)} + T_{py}^{(5)} + 2T_{py}^{(6)} \\
M_{zzzz} &= -2M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{xyyz} &= M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxz} & 0 & 0 & M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & M_{yyzx} & 0 & 0 & M_{yyxz} & 0 \\ 0 & 0 & 0 & 0 & M_{zzzx} & 0 & 0 & M_{zzxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\ M_{xxxx} & M_{xyyy} & M_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} - T_{fay}^{(2)} + 2T_{fby}^{(1)} - T_{fby}^{(2)} - 2T_{py}^{(1)} + T_{py}^{(3)} \\
M_{xxxz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} + T_{fay}^{(2)} + 2T_{fby}^{(1)} + T_{fby}^{(2)} - 2T_{py}^{(1)} - T_{py}^{(3)} \\
M_{yyzx} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} + 2T_{fay}^{(2)} - 4T_{fby}^{(1)} + T_{py}^{(3)} + 2T_{py}^{(4)} \\
M_{yyxz} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} - 2T_{fay}^{(2)} - 4T_{fby}^{(1)} - T_{py}^{(3)} - 2T_{py}^{(4)} \\
M_{zzzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} - T_{fay}^{(2)} + 2T_{fby}^{(1)} + T_{fby}^{(2)} + 2T_{py}^{(1)} + T_{py}^{(3)} \\
M_{zzxz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} + T_{fay}^{(2)} + 2T_{fby}^{(1)} - T_{fby}^{(2)} + 2T_{py}^{(1)} - T_{py}^{(3)} \\
M_{yzxy} &= M_{dxz}^{(2)} + M_{dxz}^{(5)} + 2M_{gby}^{(1)} + 4T_{fay}^{(1)} - T_{fay}^{(2)} + T_{fby}^{(2)} + T_{py}^{(1)} + T_{py}^{(4)} \\
M_{yzyx} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)} + 4T_{fay}^{(1)} + T_{fay}^{(2)} - T_{fby}^{(2)} + T_{py}^{(1)} - T_{py}^{(4)} \\
M_{xxxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} - M_{gay}^{(1)} - M_{gby}^{(1)} - 2T_{fay}^{(1)} - 2T_{fby}^{(1)} + 2T_{py}^{(1)} \\
M_{xyyy} &= M_{dxz}^{(1)} - M_{dxz}^{(3)} + 2M_{gby}^{(1)} + 4T_{fby}^{(1)} \\
M_{zzzz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} + M_{gay}^{(1)} - M_{gby}^{(1)} + 2T_{fay}^{(1)} - 2T_{fby}^{(1)} - 2T_{py}^{(1)} \\
M_{xyyz} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)} - 4T_{fay}^{(1)} - T_{fay}^{(2)} - T_{fby}^{(2)} - T_{py}^{(1)} + T_{py}^{(4)} \\
M_{xyzy} &= M_{dxz}^{(2)} + M_{dxz}^{(5)} + 2M_{gby}^{(1)} - 4T_{fay}^{(1)} + T_{fay}^{(2)} + T_{fby}^{(2)} - T_{py}^{(1)} - T_{py}^{(4)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzyx} \\ M_{xxxx} & M_{xyyy} & M_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & 0 & M_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxy} &= 2M_{dxz}^{(4)} - M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(2)} + T_{py}^{(6)} \\
M_{yzyx} &= -2M_{dxz}^{(4)} - M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} - T_{py}^{(2)} + T_{py}^{(6)} \\
M_{zxzx} &= 2M_{dxz}^{(6)} - T_{fay}^{(3)} - T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{zxyy} &= 2T_{fay}^{(3)} + T_{py}^{(5)} + 2T_{py}^{(6)} \\
M_{zzzz} &= -2M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(5)} \\
M_{xyyz} &= 2M_{dxz}^{(4)} + M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} - T_{py}^{(2)} + T_{py}^{(6)} \\
M_{xyzy} &= -2M_{dxz}^{(4)} + M_{dxz}^{(6)} - T_{fay}^{(3)} + T_{fby}^{(3)} + T_{py}^{(2)} + T_{py}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & M_{xxxx} \\
0 & 0 & 0 \\
M_{zzzx} & 0 & 0 \\
M_{yyzx} & 0 & 0 \\
0 & 0 & M_{zzxz} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
M_{zzxx} & 0 & 0 \\
0 & 0 & M_{xyyz} \\
0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzzx}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} - M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{zzzx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} - 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{yyzx} &= M_{dxz}^{(1)} + M_{dxz}^{(3)} + 2M_{gby}^{(1)} \\
M_{zzxz} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} + M_{dxz}^{(3)} + 2M_{dxz}^{(5)} + M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{zzxx} &= M_{dxz}^{(1)} + 2M_{dxz}^{(2)} - M_{dxz}^{(3)} - M_{gay}^{(1)} - M_{gby}^{(1)} \\
M_{xyyz} &= M_{dxz}^{(2)} - M_{dxz}^{(5)} + 2M_{gby}^{(1)}
\end{aligned}$$